Llewy Smith
a1706836

Assessment 1D: Big Data Analysis

Table of Contents

Llewy Smith
a1706836

Part 1: Abstract and Introduction (5 marks)

Clearly restates the background, motivation, and proposed solution.

The code and outputs referred to in this report are available via my public GitHub repository.

Abstract

This project investigates the challenge of identifying students at risk of failing their courses before the Census Date, a key early intervention point in higher education. Using the Open University Learning Analytics Dataset (OULAD), I built a pipeline that truncates temporal features at 25% of course progress, reflecting the data available to advisors at this stage. The dataset includes learning analytics, demographic, administrative, and course metadata features, with a moderately imbalanced outcome distribution (~41% fail or withdraw). To address this, I implemented feature engineering, synthetic minority oversampling (SMOTE), and recursive feature elimination (RFE). I compared Logistic Regression, Random Forest, a Multilayer Perceptron, and a Convolutional Neural Network, with performance evaluated using F1-score and PR AUC. While Logistic Regression delivered the best predictive performance, the CNN was successfully applied with LIME explainability at the individual student level. The results demonstrate both the feasibility of early identification and the importance of transparent explanations to support timely student interventions.

Introduction

Background and context

This study builds on my earlier project stages (Parts A, B and C), where the focus has been to investigate predictive modelling for student success. My project investigates whether we can identify students at risk of failing a course prior to their Census Date (approximated as 25% of the way through the course). I test this using the set of publicly available Open University Learning Analytics Datasets, known as OULAD. The input features I gather from OULAD span four broad categories:

(1) learning analytics (Virtual Learning Environment and assessments)
(2) demographic
(3) administrative
(4) course metadata

The class distribution is moderately imbalanced with 41% of courses resulting in fail or withdraw grades (which I group as failed), motivating my use of evaluation metrics (e.g. F1-score) and sampling strategies (e.g. SMOTE) that better manage this imbalance. To mimic real-world practicality, I truncate all temporal features at 25% of each course's teaching term. This ensures my model only sees what advisors would realistically know at this stage of the course. This sets the stage for evaluating predictive models under realistic early-warning conditions.

Motivation

Student retention and academic success are critical metrics for universities and are increasingly regulated in Australia. Institutions must now comply with the Higher Education Provider Guidelines (2023) and implement a Support for Students Policy tailored to their context (Australian Government Department of Education, 2023; University of Adelaide, 2023). These policies create a legislative obligation to identify students ‘at risk’ of not completing their units of study and to intervene with proactive support. Meeting this requirement demands predictive systems that are both accurate and explainable, so interventions are timely, justified, and targeted.

Early identification is especially valuable for disadvantaged groups, such as students from lower socioeconomic backgrounds or those attempting repeat enrolments, where the risk of failure is higher. However, many deep learning approaches sacrifice transparency, making them difficult - and in some cases unethical - to apply in practice. This project is motivated by the need to bridge that gap: developing predictive models that are both effective and explainable, enabling universities to meet their obligations while also improving student outcomes.

Proposed questions

The primary research question guiding this project is: Within the first 25% of teaching time, how accurately and transparently can we predict which enrolments will result in a failed course outcome?

My supporting sub-questions are:

- **Census-date early-warning accuracy:** What PR AUC, F1-score, precision and recall can be achieved with data truncated at 25% of course progress?
- **Post-census performance gain:** How much does predictive accuracy improve if data up to 50% of course progress is included, and is this gain large enough to justify waiting beyond Census Date?
- **Explainable-AI insights for intervention:** Which behavioural, demographic, and historic-grade features drive risk scores at 25% and 50%, and how can explainability methods such as LIME guide interventions?
- **Cross-dataset consistency:** How consistent are accuracy metrics and explanatory features across the OULAD, SED, and UCI datasets?

Contributions

This project makes the following contributions:

- Developed a reproducible pipeline for early-risk prediction using learning analytics and student demographic data.
- Implemented imbalanced learning and feature selection (SMOTE and RFE) to improve CNN performance under realistic conditions.
- Compared four models (Logistic Regression, Random Forest, MLP, CNN) with fair and consistent preprocessing, highlighting trade-offs in performance and interpretability.
- Delivered explainability at the individual level using LIME applied to CNN predictions, enabling actionable insights for advisors.

Part 2: Literature Review, Analysis & Visualisation (6 marks)

Provide a well-structured literature review and detailed analysis of the previous model's results with informative visualisation to support.

Comparison of Related Studies

Six studies critical to my project are expanded on below. Table 1 provides a snapshot of each - including its approaches, dataset used, results, takeaways, and relevance to my project.

Table 1: Summary of approaches, results, and relevance to project

Study	Approach & dataset	Key results & relevance
Adefemi & Mutanga (2025)	CNN–LSTM hybrid, OULAD	Achieved F1-score of 98.9%. Inspired my early models. I have concerns with reproducibility, inconsistent metrics, and target class choices. Still useful in some ways but not as a comparison.
El Jihaoui et al. (2025)	Deep regression + SHAP, Moroccan baccalaureate (340k+)	R^2 = 0.741. Early assessment was strongest predictor of success. This inspired my interpretability implementations and highlighted the importance of early assessment scores.
El Habti et al. (2025)	RF, LR, SVM, LDA on OULAD	RF achieved F1-score of 0.82. Shows that classical ML models still perform well but fell short of other studies. Suggests I could revert to traditional (often explainable) models.
Alnasyan et al. (2025)	KANFormer (attention + KAN) on OULAD	F1-score on best model = 0.945, however even their LR baseline achieved 0.936. Suggests I should try Transformer models, though results may reflect feature engineering as much as architecture.
Adnan et al. (2021)	RF and others on OULAD, multiple course checkpoints	F1-score = 0.79 at 20%, 0.88 at 60%, 0.91 at 100%. Directly supports my temporal cutoff (census-date) approach.
Zhan et al. (2024)	Privacy-preserving framework for future training datasets.	Privacy-preserving methods reduce accuracy but can be at an acceptable rate. Highlights trade-offs in my future implementation stage, consider ethics and governance alongside performance.

Adefemi & Mutanga (2025).

"A Robust Hybrid CNN-LSTM Model for Predicting Student Academic Performance"

This study was the motivation for my project's initial direction. Adefemi & Mutanga introduced a hybrid CNN-LSTM, achieving an F-score of 98.93% on the OULAD dataset. Their work demonstrates how combining convolutional and recurrent layers can capture both local patterns and temporal dependencies in educational data, an exciting direction. That said, after attempting reproduction, I have significant doubts about the reliability of their results as:

1. their reported performance is far beyond what other studies (and logical expectations) have found.
2. the results are inconsistently quoted across the paper (98.82% accuracy in one section, 98.93% in another) without explanation.
3. the paper omits crucial details on feature engineering and preprocessing steps, which are especially complex for a hybrid CNN-LSTM.
4. they evaluated using the "Pass" class as the positive target, then compared their metrics against studies that correctly used the "Fail" class or multi-class outcomes. This is equivalent to evaluating cancer detection by optimising recall for healthy patients and then claiming your model outperforms others detecting sick patients.

Despite these issues, the paper is still useful. It highlights the potential of combining CNN and LSTM architectures for educational data and points to directions worth exploring, even if its empirical claims should be treated with caution.

Alnasyan et al. (2025).

"Kanformer: an attention-enhanced deep learning model for predicting student performance in virtual learning environments"

Introducing the KANFormer model, this work fused multi-head self-attention with Kolmogorov–Arnold networks to capture complex feature interactions. The authors' goals were similar to mine. They propose that KANFormer bridges the gap between traditional ML models, which lack predictive power, and complex deep learning models, which lack interpretability. They also performed analysis on the OULAD dataset and implemented interpretability analysis using SHAP values. Comparing many models, they found that their KANFormer model outperformed all others in terms of F1-Score (but also all other metrics). Their KANFormer model achieved an F1-Score of 0.9454 on validation set data. Despite this being an impressive outcome, it is notable that their simple Logistic Regression model achieved an F1-Score of 0.9361. The authors notably conducted more extensive feature engineering than other papers, including aggregate and comparative variables. It was not mentioned whether these were made only with the training data which is important to prevent data leakage. Due to the high F1-Scores across all their models, and relatively small difference between KANFormer and Logistic regression, I suspect their success might come from their feature engineering rather than their models.

El Jihaoui et al. (2025).

"A Deep Learning Regression Model for Predicting Students' Performance with Shapley Additive Explanations-based Interpretability"

This study applied deep regression models to Moroccan baccalaureate data, incorporating SHAP and Permutation Feature Importance to identify drivers of student outcomes in the final years of high school. This dataset does not have the VLE engagement data that is present in OULAD, but has rich demographic data, course metadata, and assessment information (technically still a form of learning analytics). Its strengths are in scale (over 340,000 records) and the focus on transparency. The study interestingly found that by far the most important predictor of success for Moroccan students was their grade an early exam worth 25% of their final grade called their 'Regional Exam'. This matches up with my findings that early assessment scores are the single biggest predictor of success. Their scope differs from mine in that they are just attempting to explain the key factors, not drive actionable interventions. Despite this, some key ideas I have incorporated from their work is their breakdown of global interpretations as in Figure 9 below, from page 6 of their paper) and their local explanations (such as in Figure 14 below, from page 7 of their paper).

Fig. 9 and Fig 14 from El Jihaoui et al. (2025)

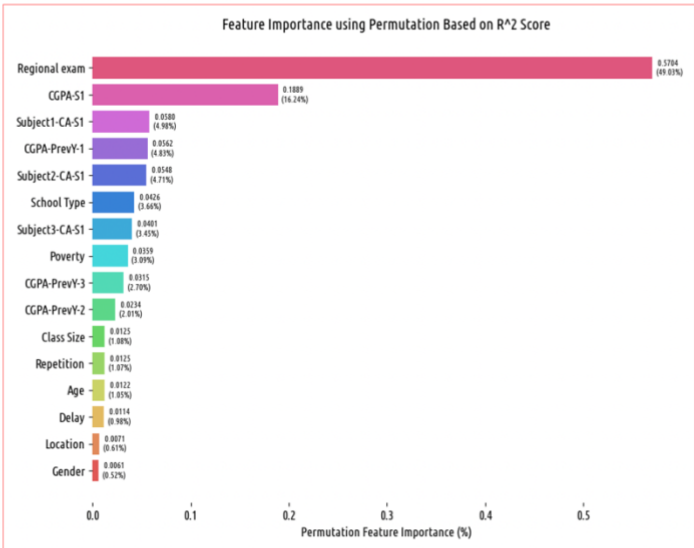


Fig. 9. Feature importance using permutation technique

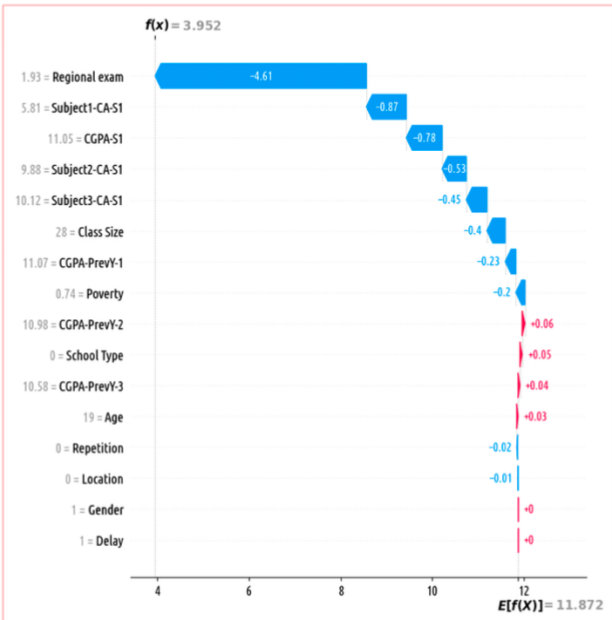


Fig. 14. Feature Contributions to Prediction for Student ID **5699**

El Jihaoui et al. (2025) achieved an MSE of 0.238 and R2 Score of 0.741. The latter means that the model explains approximately 74% of the variance in student performance in that dataset.

El Habti et al. (2025).

“Enhancing Student Performance Prediction in e-Learning Ecosystems Using Machine Learning Techniques”

This paper used relatively simple models including Logistic Regression, Random Forest, SVM, and Linear Discriminant Analysis (LDA) on the OULAD dataset. Random Forest achieved the strongest performance (91% accuracy) with others close behind. This suggests that classical ML models are still highly effective in learning analytics applications, and likely that my hybrid CNN-LSTM ambitions were not necessary. They achieved these results attempting multi-class classifications, rather than simplifying the dataset to Pass/Fail like myself and many other papers. El Habti et al.’s (2025) best model (Random Forest) generalised well to achieve a F1-Score of 0.82 on the failure class in the test dataset. It’s not a spectacular score but shows that simple and quick models can achieve very strong results in this dataset.

Adnan et al. (2021).

“Predicting at-risk students at different percentages of course length for early intervention using machine learning models”

This study evaluated OULAD predictions at multiple points (0%, 20%, 40%, 60%, 80%, 100% of course progress), finding that accuracy rises as more data accumulates. This is directly applicable to one of my research sub questions. Random Forest (RF) consistently outperformed their other classifiers. Even at 20% of the course length, their RF predictive model was achieving a 0.79 f-score, which improved to 0.88 at 60% through the course and 0.91 at 100% of the way through the course. These results set a good precedent and make this one of the most directly comparable and supportive prior works. The results are not as strong as others such as Adefemi & Mutanga’s (2025) or Alnasyan et al.’s (2025) however the study is clear, easily reproducible, and focused on developing models for timely student interventions.

Zhan et al. (2024)

“Preserving Both Privacy and Utility in Learning Analytics”

This is a study released by cross-institutional authors, one of whom is Professor Aberlardo Pardo, Head of the School of Computer and Mathematical Sciences at the University of Adelaide. Their study investigates methods that can be implemented to protect the privacy of learners while still maintaining utility (effectiveness) in the use of their learning analytics data. This work, which they describe as ‘pioneering’ is key because the end goal is not to produce models working exceptionally on OULAD students, as students at Australian institutions vary greatly, some will be like OULAD students, some will not be. Zhan et al. (2024) propose training models with privacy-preserved data (using methods from their study) and then applying these locally to learners in real-time.

Synthesis

Taken together, these studies highlight four key trends. First, strong baselines such as RF and LR remain competitive on OULAD. Second, advanced architectures (CNN-LSTM, KANFormer) push accuracy but risk opacity. Third, explainability (LIME/SHAP) is widely incorporated across studies, however implementation has not been developed to the point that interventions could be made from these results yet. Finally, timing matters: the predictive signal available at 20% of course progress is already actionable, supporting the design of this project's census-date truncation. The contribution of this project lies in combining several of these strengths:

- temporal cutoffs for timely interventions (like Adnan et al. (2021))
- model accuracy for correct interventions (like Adefemi & Mutanga (2025) and Alnasyan et al. (2025))
- individual-level interpretability (LIME/SHAP) for actionable interventions (like El Jihaoui et al. (2025))
- strong ethical considerations and privacy adherence (like Zhan et al. (2024))

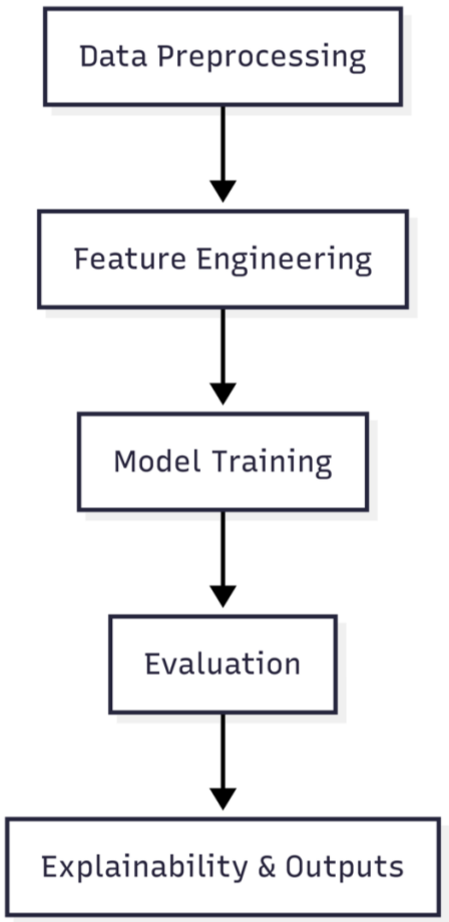
Part 3: Methodology and Improvement of Situation (5 marks)

Clearly describe the research methodology used and correctly identifies the techniques to use to find improved outcomes or models and how to use or deploy them.

Research Methodology

My methodology followed a structured pipeline of five phases.

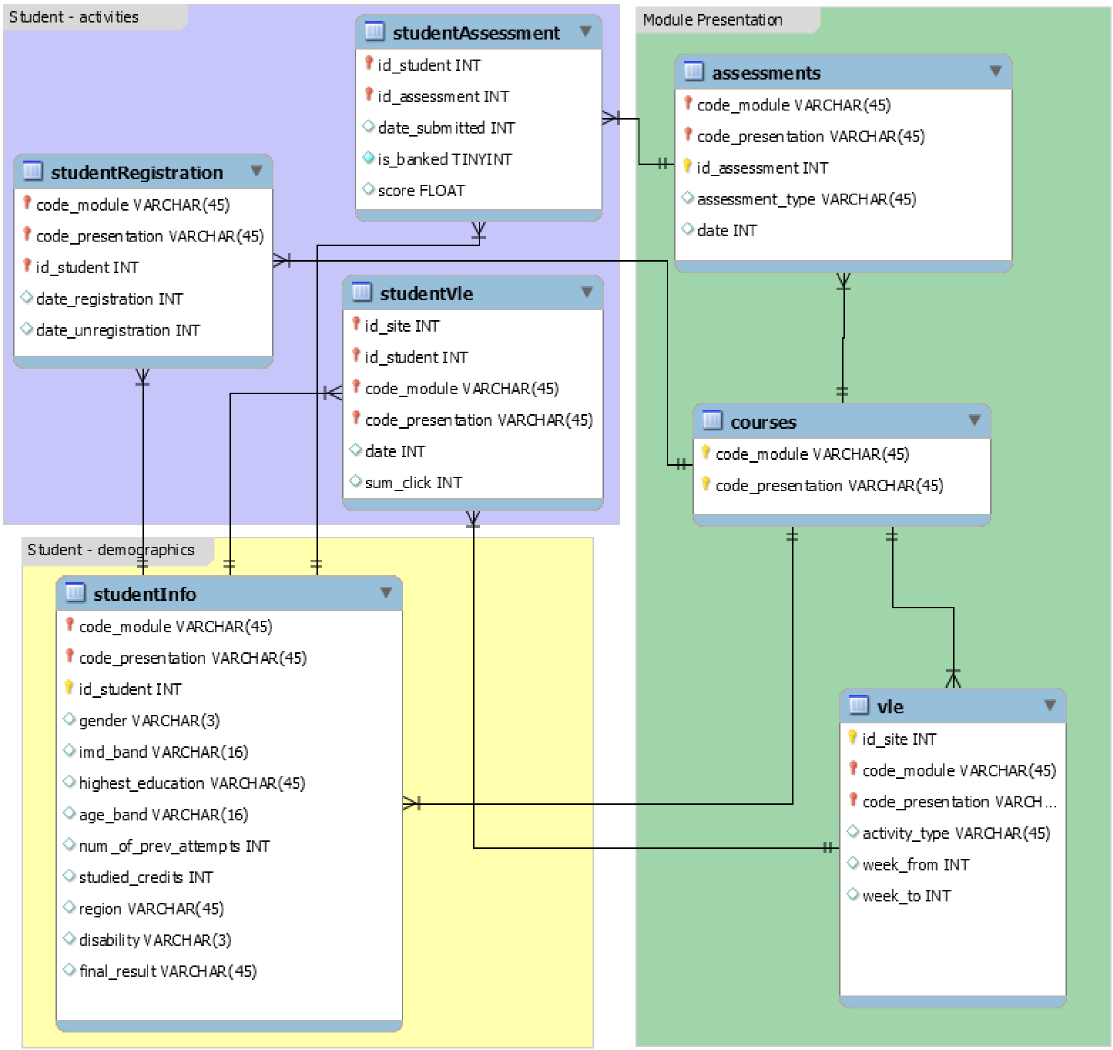
Figure 1: Methodology steps as outlined in next section



Data Preprocessing

I merged all seven OULAD tables (courses, assessments, vle, studentInfo, studentRegistration, studentAssessment, studentVLE). The full dataset schema including these tables is available publicly on the OULAD (2017) webpage, accessible here.

Figure 2: Table relations and cardinality (OULAD, 2017)



I truncated temporal features at 25% of course length to simulate Census Date conditions. I addressed Class imbalance (~41% fail/withdraw) by applying SMOTE on training data only.

Feature Engineering

Aggregate variables such as sum_click, days_clicked, and early_weighted_avg_score were created, along with binary flags (e.g., early_failed_assessment to indicate whether a student failed early assessments). The enrolment_instance_id was preserved outside the model matrix for tracking and explainability.

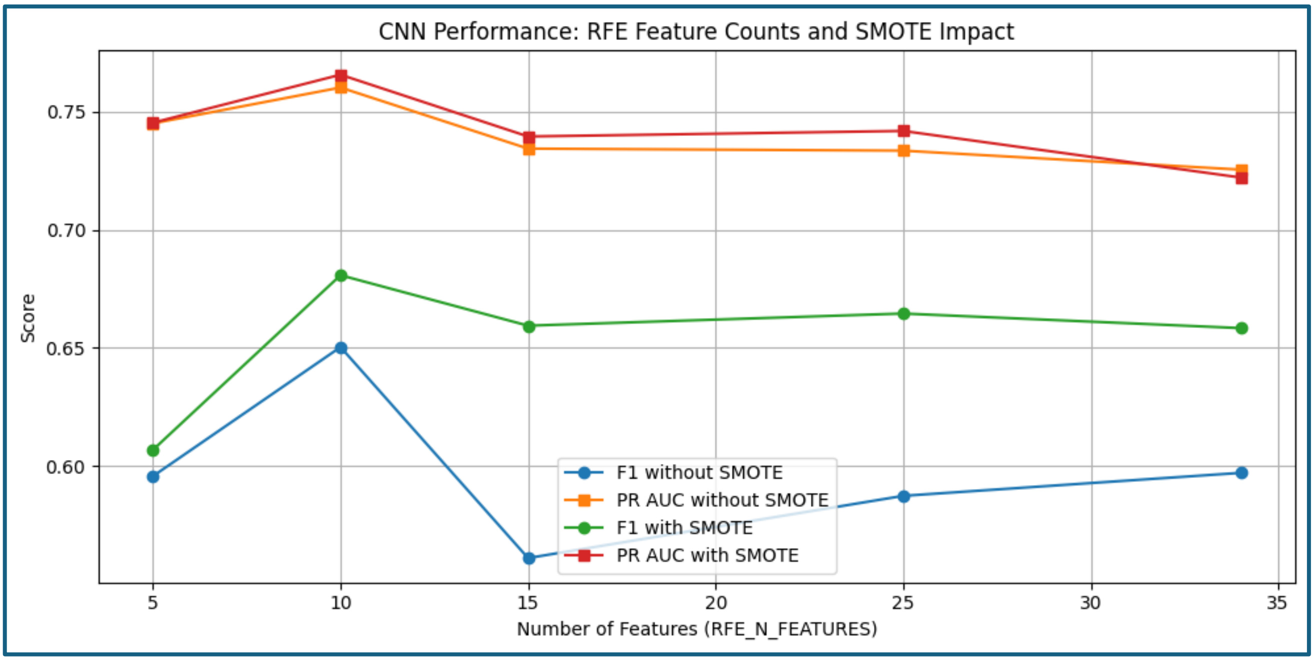
Encoding and Cleaning

A ColumnTransformer scaled numeric variables with StandardScaler() and one-hot encoded categorical variables with OneHotEncoder(drop='first', handle_unknown='ignore'). The resulting dataset was a dense, scaled input matrix with minimal missing data, where absent events were imputed as zeros.

Model Training

Four models were trained: Logistic Regression (baseline), Random Forest, Multilayer Perceptron, and a Convolutional Neural Network (CNN). The CNN architecture comprised two Conv1D layers, global max pooling, dropout (0.5), and dense layers, optimised with Adam and early stopping. Recursive Feature Elimination (RFE) was applied across feature counts {3, 5, 10, 15, 20, 25, 30, 34} for the CNN model.

Figure 3: CNN Performance: RFE Feature Counts and SMOTE Impact

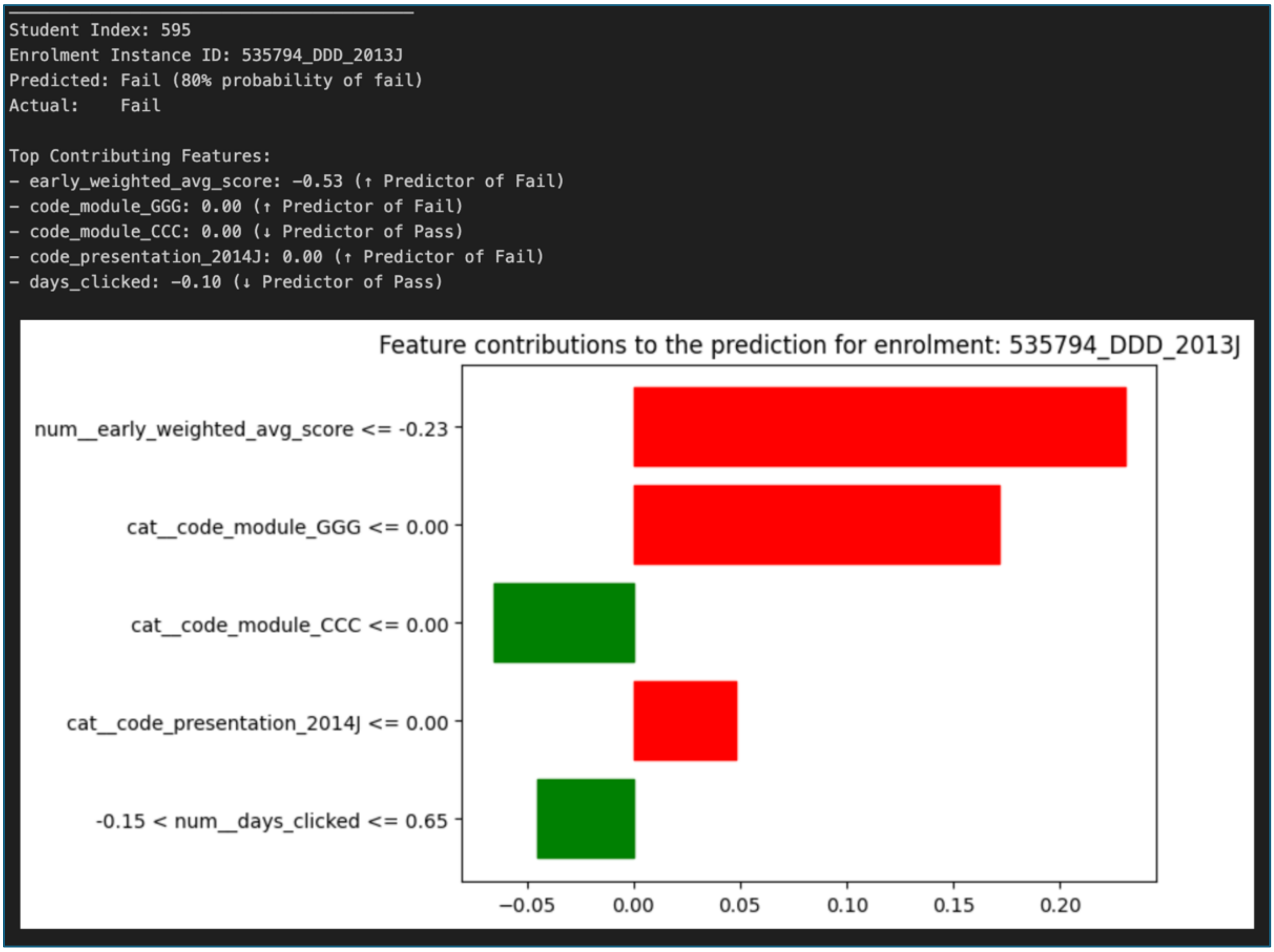


We select k=10 and apply SMOTE→ higher recall on Fail with minimal complexity; this change improved the situation over the baseline CNN without SMOTE/RFE.

Explainability and Interpretation

LIME was applied to CNN predictions at the individual student level, producing feature-based explanations (e.g., low early assessment scores, limited VLE activity) to support interpretability. Each LIME explanation shows the exact features that pushed a prediction towards ‘Fail’ or ‘Pass’ for a given student. For example, in Figure 4, one student was flagged at 80% of failing by the model. This student did indeed fail. At the 25% mark of the course, an academic advisor would have been able to use this model to (1) identify this student and their predicted level of risk and (2) see that the biggest contributory factor to this was the student’s early weighted average score of assessments, which was 0.53 standard deviations below the average.

Figure 4: LIME application on student #595



Experimental Evaluation

Experimental Setup

A stratified 80/20 train-validation split (random_state=42) was used. Evaluation metrics included F1-score (fail class), PR AUC, precision, recall, and accuracy. All experiments were run on my Mac M2 Pro using TensorFlow/Keras 3 via a Visual Studio Code Notebook (.ipynb).

Experimental Results

Logistic Regression and Random Forest achieved the strongest results (F1-score ≈ 0.68). CNN, when combined with RFE and SMOTE, reached an F1-score of 0.681 and PR AUC of 0.779 using 10 features. The Multilayer Perceptron underperformed relative to the other models.

Table 2: model performance on validation set:

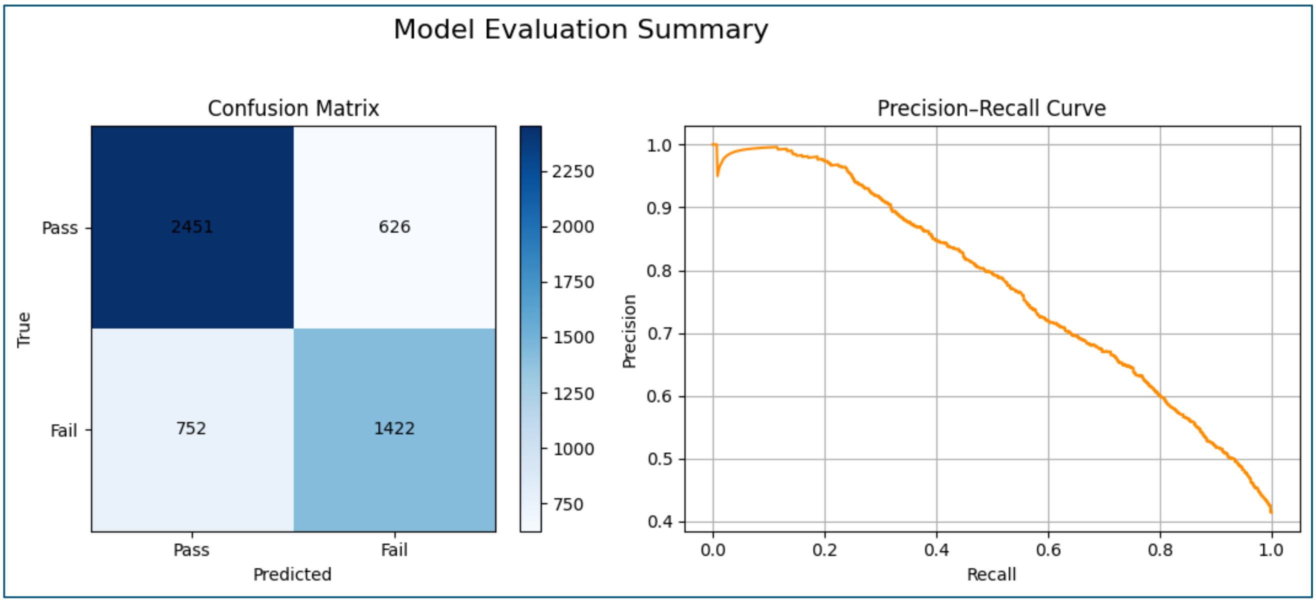
Model	Accuracy	Precision	Recall	F1-score	PR AUC
Logistic Regression	0.74	0.68	0.69	0.68	0.78
Random Forest	0.74	0.70	0.66	0.68	0.77
MLP	0.75	0.75	0.58	0.65	0.78
CNN	0.71	0.64	0.70	0.68	0.75

Key Findings

SMOTE consistently improved recall on the Fail class, raising F1-score and PR AUC. RFE simplified feature sets and improved generalisation, with 10 features giving the best trade-off. While Logistic Regression remains the simplest and most interpretable model, the CNN uniquely supports individual-level explanations through my LIME implementation.

Figure 5: CNN Model - Confusion Matrix and PR Curve

RFE: k=10, SMOTE: True



Part 4: Discussion, Conclusion and Future Work (4 marks)

Provides clear and sufficient discussion on the outcomes of the analysis results, the next direction for research and effectively concludes the report.

Discussion

- Interpret the results: key insights and takeaways
- Discuss implications for researchers and practitioners

My results show that classical models remain highly competitive on early-truncated OULAD data as was evident from the work of Adnan et al. (2021), Alnasyan et al. (2025) and El Habti et al. (2025). My Logistic Regression, Random Forest, and CNN models achieved F1-scores of 0.68 with balanced precision/recall. For the CNN, this was when paired with RFE (10 features) and SMOTE.

This aligns with prior findings from my literature review that well-engineered features plus simple models often match or close-to-match deep architectures, at least on the OULAD dataset. Practically, Logistic Regression is the preferred production model due to its simplicity, speed, and transparent coefficients. My CNN does add value for individual-level explanations via LIME that can guide advisor conversations, however I could have pursued similar with my other models such Logistic Regression or Random Forest.

The consistent lift from SMOTE underscores the importance of handling class imbalance when the fail/withdraw class sits near ~41%. Together, these results support the findings of Adnan et al. (2021) in that it is viable to make accurate pre-census interventions using data available at 25% (or 20% in their case) of course progress. My results go further to show that it is viable to also have explainable results at this early stage. For Australian Universities, an interpretable baseline model appears sufficient to operationalise early interventions with minimal engineering overhead and hence expertise required.

Limitations and Future Work

1. **Single dataset focus:** My experiments were limited to OULAD, and international and fully online university. The results generalise well to their future cohorts, but likely will not generalise to Australian cohorts and likely need local training. This is where the work of Zhan et al. (2024) becomes critical as this cannot become scalable until this ethics issue is solved.

2. **Census Date approximation:** The 25% cutoff is a proxy for Census Date and may not be accurate across institutions. It is also the data, in most cases, that withdrawals must be made by to avoid fee payments and academic impacts, meaning interventions need to be made days beforehand to give time for actions.

3. **Explainability scope:** LIME explanations were generated for the CNN however global model explanations (PFI/MDI) were not fully implemented in this final iteration. In Part B I did conduct basic Mean Decrease in Impurity (MDI), which is similar but not the same as Permutation Feature Importance (PFI) to generate global interpretations of the most

relevant features. The results gave early_weighted_avg_score (0.185), sum_click (0.137) and days_clicked (0.136) but may have changed as my feature engineering process has changed since then. In future work I would continue this and communicate it clearly like El Jihaoui et al. (2025).

4. **Overall scope:** My intended project and outcomes were beyond the remit of a single course enrolment's worth of work. A limitation was the time I was able to dedicate in this period. I have a strong plan for future work, with the eventual goal to answer all of my research questions at or above the level of current publications (of which there are many) and derive models that generalise well to Australian University datasets.

5. **Feature and model scope:** There are many ways to engineer features in this dataset to its complex hybrid structure of multiple different time-series elements and many static elements. Some researchers who focused on this, such as Alnasyan et al. (2025) saw high results across all their models. Others, such as Adefemi & Mutanga (2025) had individual complex and high-performing models. In the future I would like to pursue more testing with feature engineering, and to apply more than just 4 models, possibly including architectures such as Huang & Chen's (2024) successful temporal graph network implementation.

6. **Data leakage:** Although I took care with my feature engineering and applying RFE/SMOTE on train data only, it is still not free of indirect data leakage. As this is time series data (the same course can be taught in 2013 and 2024) I could reattempt using 2013 data for training and validation and then 2014 data for testing. This surfaces the issue of the one course that is only taught in 2014 (module CCC) but would replicate real world application with no opportunity for leakage.

Conclusion

This project demonstrates that pre-census prediction of course failure is feasible and actionable using a combination of learning analytics features available by 25% of course progress, at least in the case of the OULAD dataset. Logistic Regression provides the best accuracy-to-complexity trade-off for deployment, while a CNN with LIME has been verified to be able to effectively provide case-level insight for advising.

In my future work I will directly test my remaining sub-questions: (1) evaluating post-census gains at 50% course progress, (2) extending the pipeline to SED and UCI datasets to examine cross-dataset consistency, and (3) exploring global explainability methods (PFI/SHAP) to complement the local LIME explanations presented here.

Part 5: References

The report is written in English with good structure and logic with minimal typos or grammatical issues. Provides a comprehensive list of references. Codebase is public, clear, commented, with README.

References are progressive over Part A, B, C, and D of this report.

Datasets

Open University Learning Analytics Dataset (OULAD)

Kuzilek, J, Hlosta, M & Zdrahal, Z 2017, 'Open University Learning Analytics dataset', *Scientific Data*, vol. 4, article 170171, doi:10.1038/sdata.2017.171, viewed 22 June 2025, https://analyse.kmi.open.ac.uk/open-dataset\

Universiti Malaya Student Engagement Dataset (SED)

Kassim, MSS, Azizul, ZH & Fuaad, AAH 2025, 'Student Engagement Dataset (SED): An online learning activity dataset', *IEEE Access*, early access, viewed 22 June 2025, https://ieeexplore.ieee.org/document/10844083\

University of California-Irvine Dataset (UCI)

Cortez, P & Silva, A 2008, 'Using data mining to predict secondary school student performance', *Proceedings of the European Simulation and Modelling Conference (EUROSIS)*, EUROSIS, Ghent, pp. 117–122, viewed 22 June 2025, https://www.researchgate.net/publication/228780408_Using_data_mining_to_predict_secondary_school_student_performance/

Other References

Adnan, M, Ahmad, NH, Aziz, ME *et al.* 2021, 'Predicting at-risk students at different percentages of course length for early intervention using machine learning models', *IEEE Access*, vol. 9, pp. 7519–7539, doi:10.1109/ACCESS.2021.3050103.

Adefemi, KO & Mutanga, MB 2025, 'A robust hybrid CNN–LSTM model for predicting student academic performance', *Digital*, vol. 5, no. 2, article 16, doi:10.3390/digital5020016.

Alnasyan, B, Basheri, M, Alassafi, M *et al.* 2025, 'Kanformer: an attention-enhanced deep learning model for predicting student performance in virtual learning environments', *Social Network Analysis and Mining*, vol. 15, article 25, doi:10.1007/s13278-025-01446-7.

Australian Government Department of Education 2023, *Higher Education Provider Guidelines 2023*, Federal Register of Legislation, Canberra, viewed 22 June 2025, https://www.legislation.gov.au/F2023L00400. legislation.gov.au

Australian Government Department of Education 2024a, Success rate - Glossary, Tertiary Collection of Student Information (TCSI), viewed 22 June 2025, https://www.tcsisupport.gov.au/glossary/glossaryterm/Success%20rate.

Australian Government Department of Education 2024b, Retention rate - Glossary, Tertiary Collection of Student Information (TCSI), viewed 22 June 2025, https://www.tcsisupport.gov.au/glossary/glossaryterm/retention-rate.

El Habti, F E, Hiri, M, Chrayah, M, Bouzidi, A & Aknin, N 2025, 'Enhancing student performance prediction in e-learning ecosystems using machine learning techniques', International Journal of Information and Education Technology, vol. 15, no. 2, doi:10.18178/ijiet.2025.15.2.2243.

Huang, Q & Chen, J 2024, 'Enhancing academic performance prediction with temporal graph networks for massive open online courses', *Journal of Big Data*, vol. 11, no. 52, doi:10.1186/s40537-024-00918-5.

Jihaoui, ME, El Kheir Abra, O & Mansouri, K 2025, 'A deep learning regression model for predicting students' performance with Shapley additive explanations-based interpretability', in *Proceedings of the 5th International Conference on Innovative Research in Applied Science, Engineering and Technology (IRASET)*, Fez, Morocco, 1–8, viewed 22 June 2025, https://ieeexplore.ieee.org/document/11008069/.

Khoudi, Z, Hafidi, N, Nachaoui, M *et al.* 2025, 'New approach to enhancing student performance prediction using machine learning techniques and clickstream data in virtual learning environments', *SN Computer Science*, vol. 6, article 139, doi:10.1007/s42979-024-03622-6.

Mohammad, AS, Al-Kaltakchi, MTS, Alshehabi Al-Ani, J & Chambers, JA 2023, 'Comprehensive evaluations of student performance estimation via machine learning', *Mathematics*, vol. 11, no. 14, article 3153, doi:10.3390/math11143153.

University of Adelaide 2023, *Support for Students Policy*, University of Adelaide, Adelaide, viewed 22 June 2025, https://www.adelaide.edu.au/policies/5003/

Zhan, C, Joksimovic, S, Ladjal, D, Rakotoarivelo, T, Marshall, R & Pardo, A 2024, 'Preserving both privacy and utility in learning analytics', IEEE Transactions on Learning Technologies, vol. 17, pp. 1615–1627, doi:10.1109/TLT.2024.3393766.

Appendix 1: Abbreviations

- **EDM** - Educational data mining
- **OULAD** - Open University Learning Analytics Dataset
- **UCI** - University of California-Irvine
- **SED** - Student Engagement Dataset (SED) - Universiti Malaya
- **VLE** - Virtual Learning Environment
- **LMS** - Learning Management System
- **CNN** - Convolutional Neural Network
- **LSTM** - Long Short-Term Memory
- **PFI** - Permutation Feature Importance
- **SHAP** - Shapley Additive Explanations
- **XAI** - Explainable Artificial Intelligence
- **APP-TGN** - Attention-based Personalised Propagation - Temporal Graph Network
- **GRU** - Gated Recurrent Unit

Appendix 2: Definitions

Success Rate – The Success rate for year(x) is the proportion of actual student load (EFTSL) for units of study that are passed divided by all units of study attempted (passed + failed + withdrawn) (Department of Education 2024).

Retention rate - Retention rate is a measure of the proportion of students who continue their studies after their first year. A more detailed definition is given here (Department of Education 2024).